

ASSIGNMENT-2

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1. Drawbacks of Vacuum Tube Devices and Need for ICs

Vacuum tubes were bulky, consumed high power, produced more heat, and had low reliability. Integrated Circuits (ICs) were developed to reduce size, power consumption, cost, and improve speed and reliability.

2. Evolution of Integrated Circuits

Integrated circuits evolved from small-scale integration to very-large-scale integration.

As technology improved, more transistors were packed into smaller chips, increasing speed and reducing size.

3. Moore's Law

Moore's law states that the number of transistors on a chip approximately doubles every two years, leading to increased performance and reduced

cost.

4. States in IC Manufacturing

The main stages are:

Silicon wafer preparation

Oxidation

Photolithography

Doping

Etching

Deposition

Metallization

Packaging

5. Challenges in IC Manufacturing

Miniaturization difficulties

Heat dissipation

High fabrication cost

Defect control

6. Conversion of Silicon Oxide into

Polycrystalline Silicon

Silicon oxide is chemically reduced and purified to obtain high-purity polycrystalline silicon used for wafer

manufacturing.

7. Metallurgical Grade vs Electronic Grade Silicon

Metallurgical Grade Silicon

Lower purity

Used in metallurgy

Electronic Grade Silicon

Very high purity

Used in semiconductor devices

Purification Steps

Chemical purification

Distillation

Crystal growth

8. Conversion of Single Crystal Ingot into Silicon Wafers

The ingot is sliced into thin wafers, polished, cleaned, and prepared for IC fabrication.

9. Czochralski and Float Zone Techniques

Czochralski Method

Crystal grown from molten silicon

Suitable for large wafers

Float Zone

Method

Produces purer crystals

Difficult for large wafers due to stability issues

10. Fabrication Operations and Definitions

Main Categories

Layering

Pattern transfer

Doping

Heat treatment

Definitions

Die: Individual IC unit

Chip: Small semiconductor device

Scribe lines: Lines used for wafer cutting

11. Types of Layering Operations

Layering operations include oxidation, deposition, and epitaxy.

Layers in IC

Oxide layer

Metal layer

Polysilicon layer

Semiconductor

layer

12. Photolithography and Photoresists

Photolithography transfers patterns onto a wafer using light and masks.

Positive Resist

Exposed area becomes soluble.

Negative Resist

Exposed area becomes hard.

13. Main Doping Techniques

Thermal Diffusion

Impurities diffuse into silicon at high temperature.

Ion Implantation

High-speed ions are implanted into silicon.

Advantages

Diffusion is simple.

Ion implantation gives precise control.

14. MOSFET Fabrication Steps

MOSFET fabrication includes oxidation, lithography, doping, deposition, etching, and

metallization starting from a silicon wafer.

15. Oxidation and Oxidation Process

Oxidation is the process of forming a silicon oxide layer on a silicon wafer for insulation and protection.

16. Thermal Diffusion

Thermal diffusion is a doping process where impurities enter silicon at high temperature to change conductivity.

17. Ion Implantation

Ion implantation is a doping technique where high-energy impurity ions are forced into the semiconductor surface for precise doping.

18. Difference between Doping Techniques

Thermal Diffusion

Simple process

Less precise

Ion Implantation

Accurate doping control

Expensive

process

19. Stages of Lithography Process

Wafer cleaning

Photoresist coating

Mask alignment

Exposure to light

Developing

Etching

20. Mask and Mask Making

A mask is a patterned plate used to transfer circuit patterns onto a wafer.

Steps

Design preparation

Pattern generation

Mask

fabrication

Inspection

21. Components of Photoresist
Components

Resin

Solvent

Sensitizer

Positive Photoresist

Exposed area is removed.

Negative Photoresist

Exposed area remains after

development.

22. Conventional Lithography vs E-beam Lithography

Conventional Lithography

Uses light and masks for pattern transfer.

E-beam Lithography

Uses electron beams for very fine patterns without masks.

23. Challenges and Limitations of Wet Etching

Challenges

Poor dimensional control

Undercutting problem

Limitations

Not suitable for very small

patterns

Difficult to achieve high precision

24. Dry Etching

Dry etching removes material using plasma or reactive gases and provides better precision than wet etching.

25. Layers in Deposition Process

The main deposited layers are:

Oxide layer

Nitride layer

Polysilicon

layer

Metal layer

26. Important Film Parameters

Thickness

Uniformity

Adhesion

Stress

Electrical

properties

27. Difference between CVD and ALD
CVD

Deposits films through chemical reactions.

ALD

Deposits films layer by layer with high precision.

Advantages of ALD

Excellent thickness control

Uniform coating

Limitation

Slow